

Grade 11 Mathematics

Equations and Inequalities & Exponents and Surds

Task: Task

Term: Term 2

Time: 90 minutes

Total Marks: 50

QUESTION 1

- 1.1 Consider the equation $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.
- 1.1.1 Let $k = x^2 - 3x$. Rewrite the equation in terms of k . (1)
- 1.1.2 Solve for k . (2)
- 1.1.3 Hence, solve for x . (2)
- 1.2 Solve for x : $3x^2 + 7x > 6$ (5)
- 1.3 Solve for x : $(2x - 1)(x + 3) \geq (x + 3)(x + 5)$ (5)
- 1.4 A rectangular lawn measures 12 m by 8 m. A paved path of uniform width x metres is built all the way around the outside of the lawn. The finished paved path covers exactly the same area as the lawn.
- 1.4.1 Write down the length and the breadth of the outer rectangle (lawn and path together) in terms of x . (1)
- 1.4.2 Determine the width of the path. (4)
- 1.5 Given the equation $3x^2 - 4x - 2 = 0$.
- 1.5.1 Solve for x : $3x^2 - 4x - 2 = 0$ (correct to TWO decimal places) (3)
- 1.5.2 Hence, or otherwise, solve for x : $3(x - 1)^2 - 4(x - 1) - 2 = 0$ (correct to TWO decimal places) (2)

[25]

QUESTION 2

2.1

Given the expression $E = \frac{\sqrt{x^2-9} \cdot (x+3)^{\frac{5}{2}}}{(x-3)^{\frac{1}{2}}}$, where $x > 3$.

2.1.1 Factorise $x^2 - 9$. (1)

2.1.2 Write $\sqrt{x^2-9}$ as a product of powers of $(x+3)$ and $(x-3)$ with fractional exponents. (1)

2.1.3 Hence, simplify E to a single power of $(x+3)$. (1)

2.1.4 Use your answer to the previous part to calculate the value of E when $x=7$. (2)

2.2

For which values of x will $\frac{\sqrt{6-2x}}{x+1}$ be real? (2)

2.3

Simplify $\frac{x^2-9}{x^2-3x}$ if $x=3+\sqrt{2}$. Leave your answer in simplest surd form with a rational denominator. (5)

2.4 Solve for x , without using a calculator:

2.4.1 $3^x = \frac{1}{27}$ (1)

2.4.2 $2^{x+2} - 3 \cdot 2^x + 4^{\frac{x-1}{2}} = 96$ (4)

2.5

Given: $x = \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}}$, where the pattern of square roots continues indefinitely.

2.5.1 Write x^2 in terms of x . (2)

2.5.2 Hence, determine the value of x . (4)

2.5.3 Hence, solve for y if $2^y = x + 3$. (2)

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TOTAL [50]

MARKING GUIDELINE

QUESTION 1		
1.1.1	$k^2 - 2k - 8 = 0$	✓ substitution (1)
1.1.2	$(k-4)(k+2) = 0$	✓ factors
	$k = 4$ or $k = -2$	✓ both values (2)
1.1.3	$x^2 - 3x - 4 = 0$ or $x^2 - 3x + 2 = 0$	
	$(x-4)(x+1) = 0$ so $x = 4$ or $x = -1$	✓ first pair of x-values
	$x = 4$ or $x = -1$ or $x = 1$ or $x = 2$	✓ second pair of x-values (2)
1.2	$3x^2 + 7x - 6 > 0$	✓ standard form
	$(3x-2)(x+3) > 0$	✓ factors
	Critical values: $x = -3$ or $x = \frac{2}{3}$	✓ critical values
	$x < -3$ or $x > \frac{2}{3}$	✓✓ both parts of answer (5)
1.3	$2x^2 + 5x - 3 \geq x^2 + 8x + 15$	✓ expand

		both sides
	$x^2 - 3x - 18 \geq 0$	✓ standard form
	$(x+3)(x-6) \geq 0$	✓ factors
	$x \leq -3$ or $x \geq 6$	✓✓ answer, both parts (5)
1.4.1	Length = $12 + 2x$ and breadth = $8 + 2x$	✓ both dimensions (1)
1.4.2	$(12 + 2x)(8 + 2x) - 96 = 96$	✓ setup
	$4x^2 + 40x - 96 = 0$, so $x^2 + 10x - 24 = 0$	✓ standard form
	$(x+12)(x-2) = 0$	✓ factors
	$x \neq -12$ (a width cannot be negative), so the path is 2 m wide.	✓ answer with rejection (4)
1.5.1	$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-2)}}{2(3)}$	✓ substitution
	$x = 1,72$ or $x = -0,39$	✓✓ both answers (3)
1.5.2	$x - 1 = \frac{4 + \sqrt{40}}{6}$ or $x - 1 = \frac{4 - \sqrt{40}}{6}$	✓ recognising $x - 1$ as the root

	$x=2,72$ or $x=0,61$	✓ both answers (2)

QUESTION 2		
2.1.1	$(x+3)(x-3)$	✓ factors (1)
2.1.2	$(x+3)^{\frac{1}{2}}(x-3)^{\frac{1}{2}}$	✓ fractional exponents (1)
2.1.3	$E=(x+3)^3$	✓ combine exponents (1)
2.1.4	$E=(7+3)^3$	✓ substitution
	$E=1000$	✓ answer (2)
2.2	$6-2x \geq 0$, so $x \leq 3$	✓ radicand non-negative
	$x \leq 3$ and $x \neq -1$	✓ denominator non-zero (2)
2.3	$x^2-9=(x-3)(x+3)$	✓ factorise

		numerator
	$x^2 - 3x = x(x-3), \text{ so } \frac{(x-3)(x+3)}{x(x-3)} = \frac{x+3}{x}$	✓ factorise denominator and simplify
	$\frac{x+3}{x} = \frac{6+\sqrt{2}}{3+\sqrt{2}}$	✓ substitution
	$\frac{6+\sqrt{2}}{3+\sqrt{2}} \cdot \frac{3-\sqrt{2}}{3-\sqrt{2}} = \frac{18-6\sqrt{2}+3\sqrt{2}-2}{9-2}$	✓ rationalise denominator
	$\frac{16-3\sqrt{2}}{7}$	✓ answer (5)
2.4.1	$3^x = 3^{-3} \text{ so } x = -3$	✓ answer (1)
2.4.2	$2^{x+2} - 3 \cdot 2^x + 2^{x-1} = 96 \text{ so } 2^{x-1}(2^3 - 3 \cdot 2 + 1) = 96$	✓ rewrite base and factorise lowest power
	$3 \cdot 2^{x-1} = 96$	✓ simplify coefficient
	$2^{x-1} = 32 = 2^5$	✓ same base
	$x - 1 = 5 \text{ so } x = 6$	✓ answer (4)
2.5.1	$x^2 = 20 + \sqrt{20 + \sqrt{20 + \dots}}$ and $\sqrt{20 + \sqrt{20 + \dots}} = x$	✓ recognise recursive structure
	$x^2 = 20 + x$	✓ answer (2)
2.5.2	$x^2 - x - 20 = 0$	✓ standard

		form
	$(x-5)(x+4)=0$	✓ factors
	$x=5$ or $x=-4$	✓ roots
	Since $x>0$, reject $x=-4$: $x=5$	✓ rejection of negative root (4)
2.5.3	$2^y=8=2^3$	✓ substitution and same base
	$y=3$	✓ answer (2)

TOTAL [50]

COGNITIVE LEVEL ANALYSIS GRID

QUESTION	Knowledge	Routine	Complex	Problem-Solving	Total
QUESTION 1	1	13	7	4	25
QUESTION 2	5	9	7	4	25
TOTAL	6	22	14	8	50
FET TARGET %	20%	35%	30%	15%	—
THIS TASK %	12%	44%	28%	16%	100%

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$